

FEEL TECH Wear : Enhancing Mixed Reality Experience with Wrist to Finger Haptic Attribution

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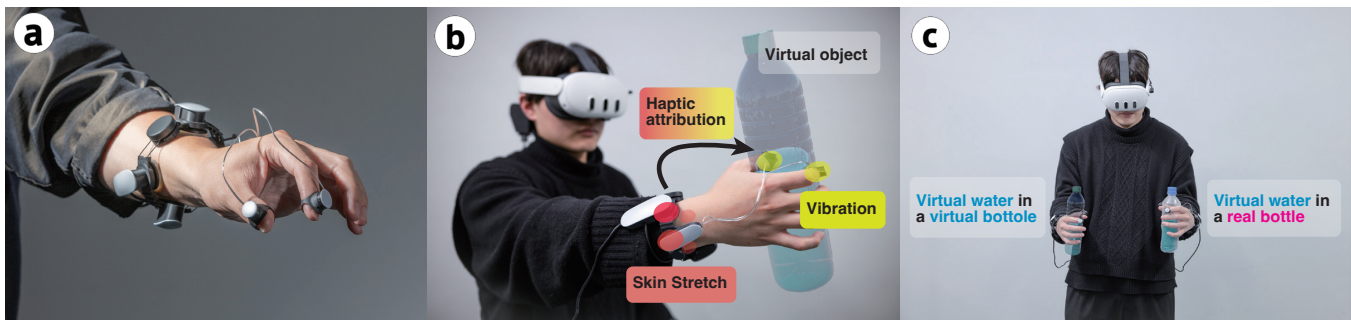


Figure 1: (a)The user is wearing our haptic device. (b)Haptic sensation is attributed to the fingertips. (c) The user can interact with virtual objects through both virtual and real objects.

ABSTRACT

FEEL TECH Wear is a system that facilitates haptic interactions while keeping most of the palm free, by presenting directional force through rotational skin-stretch distribution feedback to the wrist and providing texture sensation through vibration feedback to the fingertips. With advancements in hand tracking and passthrough technologies, hand interactions in Mixed Reality (MR) environments have become more accessible, necessitating palm-free haptic feedback methods that do not hinder interactions with real objects or impair vision-based hand tracking. The hardware of FEEL TECH

Wear primarily consists of two components: a hand-mounted device for each hand and a control unit located at the back of the head. The hand-mounted device is equipped with four channels of rotational skin-stretch factors at the wrist and vibration factors at the thumb and index finger. Using FEEL TECH Wear, three applications have been realized: haptic feedback for virtual objects, haptic augmentation for real objects, and haptic guidance towards objects.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality; Haptic devices.

KEYWORDS

mixed reality, haptics, skin-stretch, vibration, haptic attribution, palm free

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1 INTRODUCTION

In Mixed Reality (MR) environments, palm-free haptic feedback is preferable to avoid interference with real object interactions. While haptic feedback devices for Virtual Reality (VR), such as gloves that cover the hands, are accepted, MR requires that hand interactions with real objects are not hindered, setting a design guideline for these experiences. To achieve haptic experiences without obstructing hand use, research is progressing on presenting haptic feedback to different parts of the body than the original sites of perception, enabling palm-free haptic feedback. It has been revealed that the appropriate location for presenting reflection force felt by the index finger is on the palm side of the wrist[Moriyama et al. 2018]. Systems have been proposed that use bracelet devices to present squeeze and vibration feedback, facilitating various interactions such as push, pull, and rotate[Pezent et al. 2022]. However, systems that present stimuli to different parts than the original sites of perception can cause discomfort due to the difference in the location of presentation compared to direct haptic feedback to the fingertips.

Therefore, we propose a system named FEEL TECH Wear, which allows for haptic feedback to be maintained palm-free by attributing the sense of touch to the fingertips through force feedback at the wrist and vibration feedback at the fingertips. FEEL TECH Wear is capable of seamlessly presenting haptic feedback during interactions with virtual objects in Mixed Reality spaces, as well as during interactions with virtual objects through real objects, making it a versatile system for enhancing MR/AR experiences.

2 FEEL TECH WEAR

The FEEL TECH Wear hardware comprises two main components: 1.) A hand-mounted device featuring 4-channel rotational skin-stretch tactors on each wrist and vibration tactors on the thumbs and index fingers, and 2.) A control unit positioned at the back of the head. Inspired by conventional haptic devices[Horie et al. 2023], the hand-mounted device utilizes small serial servos for angle control. The ring-shaped vibration tactors are equipped with LRAs (Linear Resonant Actuators) and are capable of producing strong vibrations despite their small and lightweight design. All actuators are independently controllable. The unit at the back of the head is equipped with a microcontroller to control actuators. In order to maintain a lightweight and compact form factor for FEEL TECH Wear while ensuring stable operation, the control and power paths were carefully considered. The use of differential amplifiers for the tactile vibration signal maintains high noise resistance in the harness. Furthermore, the power supply for the servos is stepped down to the driving voltage within the wrist device using a regulator, minimizing voltage drops.

The software employs the Unity game engine to render visual and haptic virtual objects create a hybrid environment that allows

seamless transition between MR and VR environments in real time. Vision-based hand tracking is employed, and upon detecting contact with a virtual object, two patterns of haptic rendering are performed:

- (1) The combined force applied to the fingertips and the wrist is decomposed orthogonally in the wrist's transform-based vector space, and the rotation angle of the tactors is calculated from the coefficients of each basis.
- (2) The vibration at the fingertips is generated based on the texture of the contact object and the relative speed between the fingertips and the virtual object.

3 APPLICATIONS

At SIGGRAPH 2024 Emerging Technologies, participants will wear head-mounted displays and haptic feedback devices to engage in three types of hand interactions with real and virtual objects:

- (1) Presentation of the reaction force, weight and texture sensations of the virtual objects
- (2) Augmentation of the weight and texture sensation of physical objects in real space
- (3) Guidance to the necessary object locations through the presentation of directional force

In interaction (1), the reaction force of virtual objects is calculated using the penalty method, and the inertia of grab objects is calculated from acceleration of the virtual objects. The texture sensation is presented to the fingertips by calculating the gain and pitch and stretch of vibration based on the relative speed between contacted objects and the fingertips.

In interaction (2), the augmentation of the weight sensation of physical objects in real space is performed by four channels of rotational skin-stretch tactors at the wrist, and the augmentation of texture sensation is performed by vibration tactors at the fingertips. The haptic rendering techniques used are similar to those employed for presenting the reaction force, weight and texture sensation of virtual objects.

In interaction (3), the user's hand is guided to the location of objects necessary for interaction through the directional force presented by the rotational skin-stretch distribution provided by the four channels of rotational skin-stretch tactors at the wrist.

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